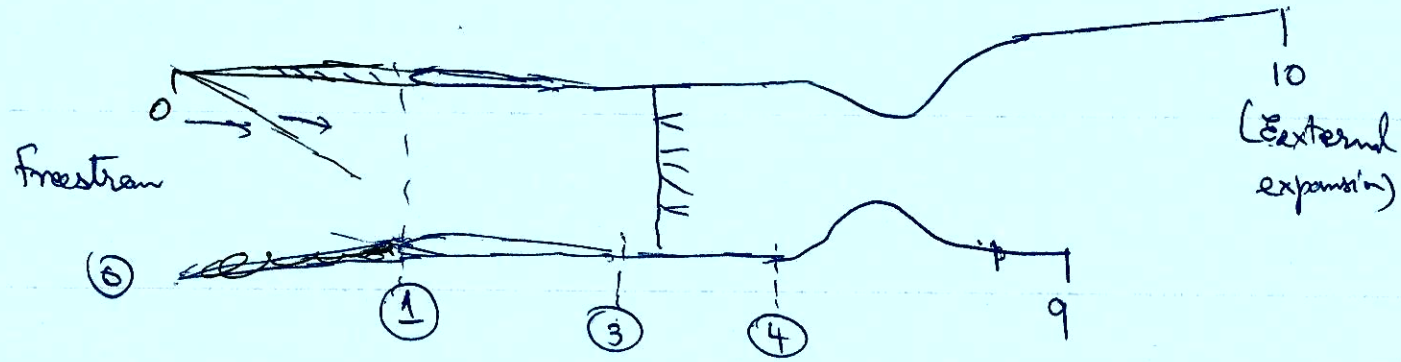
Chapter 4:

Look @ performance of the engine

Ramjets



$0 \to 1$: external compression, adiabatic, not isentropic

$1 \to 3$: Internal compression, adiabatic, possibly isentropic

$3 \to 4$: Combustor (isobaric or isochoric)

$4 \to 9$: Internal expansion, adiabatic, possibly isentropic

$9 \to 10$: External expansion, isentropic (if $p_9 > p_0$)

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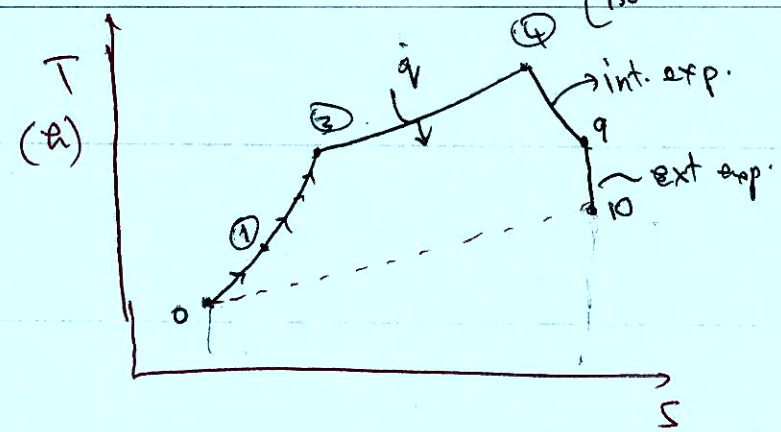
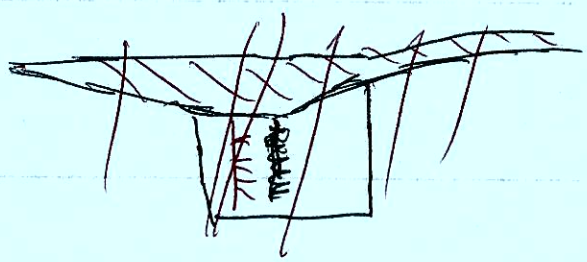
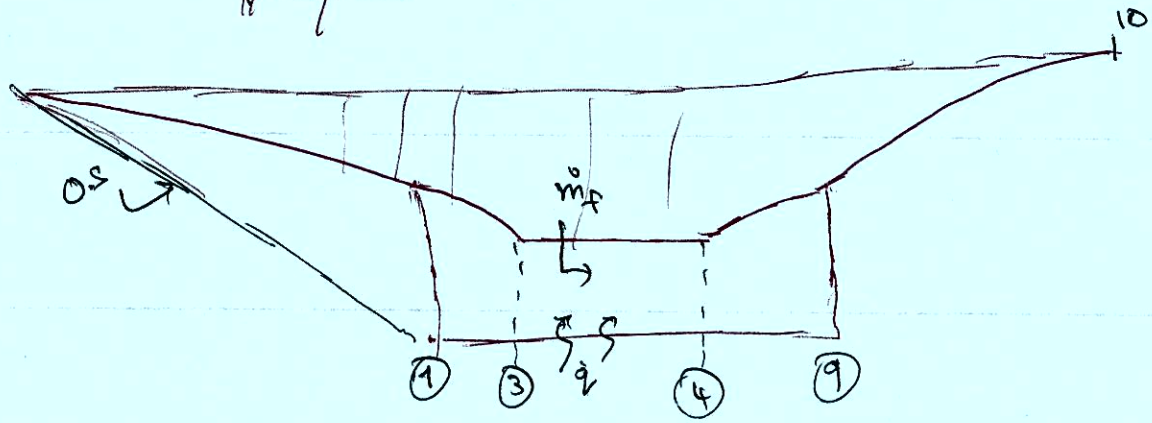
On a T-s diagram



For scramjet.





On a T-s diagram, ramjets & scramjets look the same (along w/ turbojet). In H-K space, they look different.

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Look @ entropy generation $0 \to 1$, we can show

$$\Delta s = C_v \ln\left[(1+\beta)\left(1-\beta\left(\gamma+\frac{\gamma+1}{\gamma}\beta\right)^{-1}\right)^{\gamma}\right]$$

β : shock strength $= \frac{2\gamma}{\gamma+1}(M^2-1)$ $\quad$ $\left(\text{Note } \frac{p_2-p_1}{p_1} = \frac{2\gamma}{\gamma+1}(M_1^2-1)\right]$ for N.S.

For $M \leq 1.3$, Δs is small & compression can be treated as isentropic (good). We want to design external compression or ramjet (or aerospikes) to yield $M_1 \approx 1.3$

For $M > 1.3$, can show that

$$\frac{\Delta s}{R} \propto \frac{2}{3}\frac{\gamma}{(\gamma+1)^2}(M^2-1)^3$$

Δs can increase very quickly!

Now, return to thermodynamic efficiency.

$$\eta_{th} \equiv \frac{\text{nett work}}{\text{heat in}}$$

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1st. law

$$\delta q - \delta w = \Delta h + \Delta ke + \Delta pe.$$

For a cycle.

$$RHS = 0$$

$$\delta q = \delta w$$

$$q_{in} - q_{out} = W_{out} - W_{in}$$

$$W_{in} = 0$$

$$W_{out} = q_{in} - q_{out}$$

$$\eta_{th} = \frac{W_{out}}{q_{in}} = 1 - \frac{q_{out}}{q_{in}}$$

For 3 − 4, $W = 0$

$$Q_{in} = \dot{m}_f H_{vf} = \dot{m}_a q_{in} = \dot{m}_a (h_4 - h_3)$$

$$(\Delta KE \,\&\, \Delta PE = 0)$$

$$q_{in} = f H_{vf} = \eta_b f H_{vf}.$$

η_b : burner efficiency.

Want to determine q_{out}. To do this, need to apply 1st law for $10 \to 0$

$$q_{10-0} - \underbrace{\cancel{W_{10-0}}}_{0} = \Delta h + \underbrace{\cancel{\Delta KE}}_{\approx 0} + \cancel{\Delta PE}^{\,0}$$

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$$q_{10-0} = \Delta h = q_{out} = h_{10} - h_0$$

$$\eta_{th} = 1 - \frac{q_{out}}{q_{in}}$$

$$\eta_{th} = 1 - \frac{h_{10} - h_0}{\eta_b f H_{vf}}$$

For CPG

$$\eta_{th} = 1 - \frac{c_p (T_{10} - T_0)}{\eta_b f H_{vf}}$$

<u>2nd Law</u>

$$ds \geq \frac{\delta q}{T}$$

For reversible process,

$$\delta q = T ds$$

$$q_{in} = \int_3^4 \delta q \cong \int_3^4 T ds$$

$$q_{out} = -\int_{10}^{0} \delta q \cong -\int_{10}^{0} T ds = \int_0^{10} T ds$$

$q_{net} = q_{in} - q_{out} \Rightarrow q_{net}$ is the area in T-S-space. & also net work is the area in p-v space } for a cycle these are equal ($\Delta E = 0$)

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η_{th} scales w/ pressure @ which heat is added.

Ideal Brayton cycle

$$\eta_{th} = 1 - \left(\frac{p_3}{p_0}\right)^{-\left(\frac{\gamma-1}{\gamma}\right)}$$

So, for max η_{th}, want high p_3.

However T_3 scales with p_3.

$$T_3 = T_0 \left(\frac{p_3}{p_0}\right)^{\frac{\gamma-1}{\gamma}}$$ for isentropic $0 \rightarrow 3$ compression.

$$\Rightarrow T_3 \propto p_3^{0.29}$$

T_4 is limited to $\simeq 3000-3500K$ due to dissociation.

& Work produced is proportion to heat added.

So, we want T_3 to be sufficiently low w.r.t. T_4 so that we have large q_{in} ($= W_{out}$), but want T_3 to be as high as possible for thermodynamic efficiency (η_{th})

Compromise & choose $T_3 = 1400-1700K$ to produce good η_{th} & work (ΔKE)

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So T_3 dictates the burner inlet Mach # M_3.

$$T_{t0} = T_{t3} \quad \text{(adiabatic.}$$

$$T_t = T\left(1 + \frac{\gamma-1}{2} M^2\right)$$

$$T_{t0} = T_0 \left(1 + \frac{\gamma-1}{2} M_0^2\right)$$

$$T_{t3} = T_3 \left(1 + \frac{\gamma-1}{2} M_3^2\right)$$

$$\Rightarrow T_0\left(1 + \frac{\gamma-1}{2} M_0^2\right) = T_3 \left(1 + \frac{\gamma-1}{2} M_3^2\right)$$

$$M_3 = \left\{ \frac{2}{\gamma-1} \cdot \left(\frac{T_0}{T_3} \left(1 + \frac{\gamma-1}{2} M_0^2\right) - 1 \right) \right\}^{1/2}$$

$$M_0 = \left\{ \frac{2}{\gamma-1} \left(\frac{T_3}{T_0} \left(1 + \frac{\gamma-1}{2} M_3^2\right) - 1 \right) \right\}^{1/2}$$

For ideal ramjet $M_3 \to 0$

$$T_3 = T_{t3} = T_{t0}$$

$$\psi = \frac{T_3}{T_0} \Rightarrow M_0 \text{ for } M_3 = 0 \Rightarrow M_{0,min}$$

$$M_{0,min} = \left\{ \frac{2}{\gamma-1} \cdot (\psi - 1) \right\}^{1/2}$$

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For $\psi = \frac{1550}{225} \cong 7$

$$M_{0,min} = 5.5$$

Other critical condition is when $M_3 = 1$

$$M_{0,M_3=1} = \left\{ \frac{2}{\gamma-1}(\psi-1) \right\}^{1/2}$$

For e.g., $\psi = 7$, $\gamma = 1.4$

$$M_{0,M_3=1} = 6$$

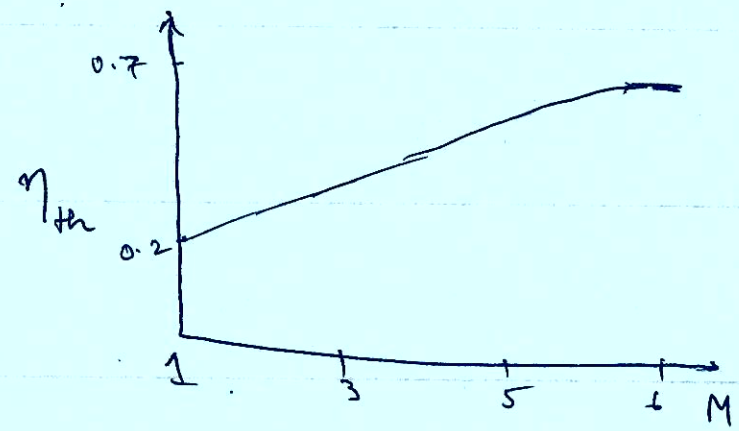
So for ramjets $(M_3 \cong 0)$

$$1.0 \le M_0 < 5.5 \quad \text{(ideal ramjet)}$$

What happens when $M_0 < 5.5$?

As $M_0 \downarrow$, $T_3 \downarrow$ (assuming $M_3 = 0$) $\Rightarrow \psi \downarrow$

$\Rightarrow \eta_{th} \downarrow$



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To get $M_0 > 6$, need to let $\underbrace{M_3 > 1}_{\text{scramjet}}$.

For scramjets $M_3 > 1$

$$6 \leq M_0 < ?$$

M_0 is limited by ability to burn fuel & oxidizer into saturated comb. products

→ mixing problem

→ kinetics problem

Generally for HC fuel, $M_3 < 2.5$
for H_2 fuel, $M_3 < 4$ } due to kinetics vs. residence time.

Solving for M_0, using M_3 for HC & H_2.

M_0 for HC $\approx$ 8-9

for H_2 $\approx$ 12-15

To go faster.

a) increase limit on T_3 (difficult. b/c pr is low, T_3 is hi ⇒ dissociation)

b) Increase rate of mixing & burning (increases M_3 limit ; also difficult)

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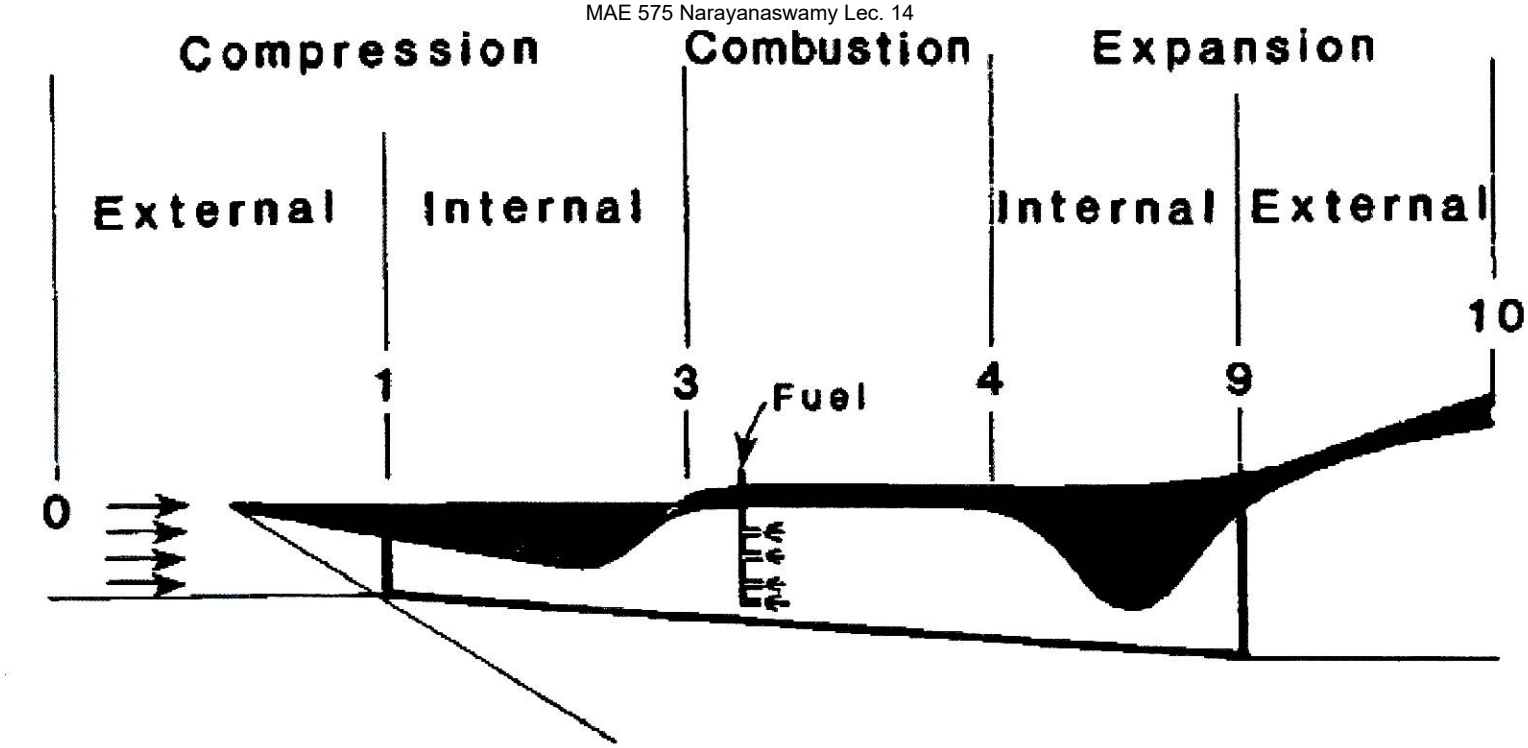
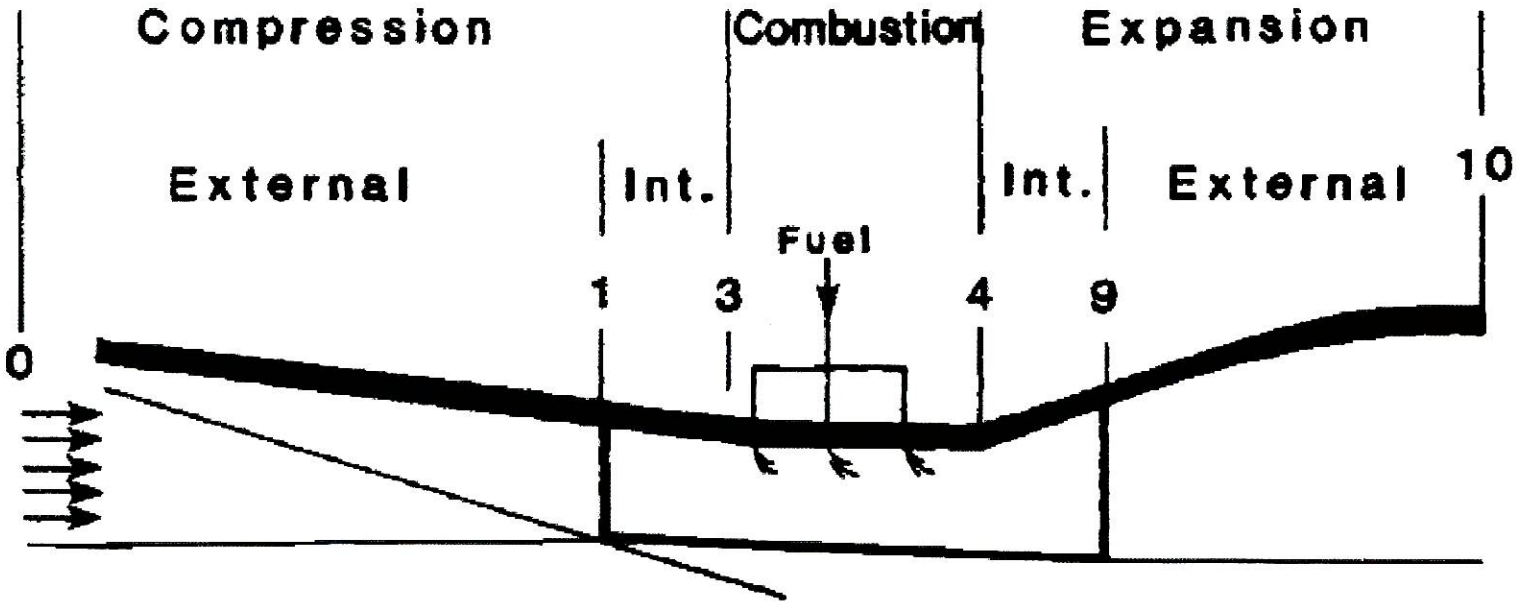


Ramjet



Scramjet

. 4.1 Airbreathing engine reference station numbers and
ed terminology.